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What is a game review: A case study approach to defining player reviews

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ABSTRACT

Online game reviews are a category of customer feedback for video games that contain valuable information for either a game's development team or its potential players. Currently, Steam is the most popular video game distribution platform, encouraging players to provide feedback by leaving reviews on the store page. However, little research has been conducted to determine the structures and characteristics of these user reviews on Steam. This paper aims to identify the types of information contained within these reviews, as well as how this information is structured. It takes the game *No Man's Sky* as a case study and employs both qualitative textual analysis and the latent Dirichlet allocation topic model method. The contribution of this study is to propose a baseline model of a game review by identifying their generalisable characteristics to support better natural language analysis of Steam game reviews. Our results show that game reviews on Steam are characterised by their capacity to include mixed information. The review data analysis should account for this diversity of meaning to accurately summarise players' views.

1. Introduction

Game reviews play an important role in the video games industry. For game producers, reviews provide potentially valuable customer feedback [1,2], enabling developers to evaluate players' perceptions [3], acceptance [4], and satisfaction [5] in meaningful ways. Here, 'player' refers to the general player who may post reviews as a form of customer feedback, rather than being a collective term for all platform users, which may also include experts. In this paper, a 'review' is defined exclusively as a player-generated report appearing on the store website (Steam, in this study). Reviews written by experts or organisations and published on Steam Curator[1] are not covered in this research.

Previously, most studies have focused on expert-generated reviews [6]. While some suggest that such reviews provide a valuable resource [7], others have questioned their validity and value [8,9]. Consequently, investigating general player-generated reviews is becoming increasingly prominent in video game studies.

Natural language processing (NLP) technology has been introduced to analyse the large volumes of player-generated textual data available within game reviews [10,11]. However, the performance of NLP techniques in processing such data remains unsatisfactory. Viggiato et al. [12] reported that applying sentiment analysis to games reviews, adopting Steam's recommendation labels as sentiment labels, might not perform effectively, despite this approach being a mature and accepted NLP technique for other corpora. The reasons for this are diverse [12,13] and related to a fundamental question: What are the features of a game review?

The purpose of this study is to investigate the nature of player-generated game reviews, focussing on the defining forms and content types that distinguish such data from other text corpora. Understanding the nature of a review will enable more accurate and detailed machine extraction of information that may be useful to developers as they seek to improve and update games. This initial study uses player reviews of *No Man's Sky* [14] on the Steam platform as a case study; future research will explore a wider range of game reviews.

The contributions of this study include the identification of key characteristics of Steam game reviews, such as formal, lexical, and linguistic components. Secondly, identifying review features on Steam improves data extraction over simpler annotated datasets, enhancing the application of numerical metrics and machine learning algorithms. This study aims to evaluate the accuracy of contextual information extracted using both qualitative analysis methods and NLP-based topic modelling, a powerful statistical approach to identify topics within document collections [15]. Finally, our case study approach enables a detailed examination of the contextual data of video games, enhances the performance of NLP models, and provides valuable insight into the outcomes of unsupervised machine learning.

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[1] https://store.steampowered.com/curators/topcurators/

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2. Related work

Previous research has used manual methods, such as coding and reading, to explore various aspects of reviews, including gender representation [16], cultural differences [1], and race representation [17]. However, with the increasing size of datasets, game researchers have begun applying text mining and data mining techniques to analyse game reviews.

2.1. Topic modelling analysis in game reviews

Topic modelling has become commonplace, particularly for handling relatively short content [18]. According to Lazreg [19], topics can be identified based on groups of words within the topic model. In recent years, researchers have increasingly used latent Dirichlet allocation (LDA), a probabilistic model that effectively infers topics from documents, to analyse player-generated game reviews. For example, Kwak et al. [20] applied LDA to investigate the relationship between player addiction and game success. Similarly, combining sentiment analysis and LDA, Li et al. [21] calculated a playability score from game reviews. Yu et al. [22] used LDA to identify esports players' opinions based on Steam reviews.

However, these studies did not provide a convincing explanation of how topics were inferred. The LDA topic model can be ambiguous, as the final inferred topics may be prone to overinterpretation. Fortunately, this subjective effect can be mitigated by quantitative and qualitative verification methods [18]. Specifically, quantitative methods, such as regularised regression, help to determine relationships between variables and inferred topics, while qualitative methods derive contextual information from the data [23].

Structural Topic Models (STM) such as those proposed by Roberts et al. [24], based on a traditional probabilistic-based topic model, but incorporating arbitrary metadata into the model. Lu et al. [4] used STM to investigate changes over time on various topics related to a specific game on Steam. Busurkina et al. [25] used STM to cluster 120 topics from game reviews, reflecting the different dimensions of the game experience. At a superficial level, the output of STM yields more topics than LDA. Reviewing the results of Lu et al. [4] and Li et al. [21], as both investigated the game *No Man's Sky*, the main categories of STM and LDA are similar. Unlike LDA,the data preprocessing of STM relies on the textProcessor[2] package, which processes raw data in a quick but straightforward form, a peculiarity that might cause the accuracy of the word input of the model to be questioned.

2.2. Game reviews analysis on steam

As one of the largest and most popular online video game distribution platforms for PC gaming [26], Steam allows (and encourages) players to write reviews about the games they purchase [25]. These reviews do not have length restrictions [17]. Each game allows only one review per user, although users can edit their review with each game update [3]. Additionally, Steam actively encourages game owners to indicate whether they would recommend the game to others. Reviews are marked as 'Recommended' or 'Not Recommended', and Steam uses these results to calculate positive or negative ratings for games [27].

Many previous studies have used the recommendation flag as the sentiment label to train their sentiment classification models [27,28]. However, researchers have overlooked evaluating the extent to which the recommendation flag equates with the sentiment polarity of the Steam platform before initiating the sentiment classification tasks. Moreover, given the linguistics of natural language, this inference did not account neutral classification. Hence, the conceptualisation of specific features in Steam remains ambiguous and unstandardised for academic research.

Lin et al. [3] conducted an empirical study that analysed 6,224 game reviews on Steam, manually categorising and labelling each review's topic. Kosmopoulos et al. [27] proposed a framework to summarise game reviews on Steam that identifies and combines the topic with sentiment analysis. Being the first systematic academic summarisation of Steam reviews, their study was inevitably exploratory. It might have been more beneficial if this system controlled the game genres, in addition to being able to assess a particular gaming context of a game or games with similar genres. In this case, the contextual information had been well considered and would have been more authoritative.

Despite the promise of NLP and machine learning algorithms, several significant issues remain outstanding. First, the nature of the language of Steam game reviews is neither consistent nor similar to that of other corpora. The existing game-specific terminology and jargon was found to negatively impact the precision of game review analysis [29]. Meanwhile, sarcasm or irony present in game reviews introduces challenges to applying automated methods, reflecting the nuanced and subjective nature of online content. Phillips et al. [30] employed the LDA approach to analyse thematic patterns in Steam game reviews. Their work highlights the importance of accounting for nonliteral language, such as sarcasm, as it adds complexity and context-dependent nuances to game reviews.

Finally, unlike the summary formats of professional reviews, Steam user reviews usually discuss multiple topics within a single review [31,32]. With various aspects corresponding to different emotions, one review may include multiple topics and viewpoints. Certain studies have sought to adopt aspect-based sentiment analysis (ABSA) to resolve this complexity [33,34]. However, this method lacks an evaluation benchmark dataset for the video game review domain, which impedes the construction of standardised annotations.

3. Method

In order to investigate the characteristics of game reviews, we carry out a case study following a mixed methods approach to identify the features of Steam reviews.

3.1. Data collection

In 2016, Hello Games[3] released the survival game *No Man's Sky*[4] (NMS). It has four primary activities: exploration, survival, combat, and trading. The game has demonstrated longevity and has many followers and online reviews. To satisfy players, the development team repeatedly overhauled various aspects of the game and subsequently uploaded. The number of major updates ranged from two to three annually from 2016 to 2019 increasing to five or six annually during 2020 and 2021. Active community support motivates the diversity and uniqueness of topics in its reviews.

The open application programming interface on Steam allows researchers to access data on Steam [8]. A custom crawler was used to get a snapshot of the game reviews available in the NMS Steam Community. In total, the programme sytematically collected 21,000 English reviews from October 2016 to October 2019 together with associated data including recommendation flags, playing hours, the number of helpful, and the number of funny reviews. This study follows ethical guidelines [35] for researching data from social networks. Any information that could violate user privacy remained uncollected and unused. Table 1 provides a summary of the data collected for each review.

[2] https://www.rdocumentation.org/packages/stm/versions/1.3.7/topics/textProcessor

[3] https://hellogames.org/

[4] https://www.nomanssky.com/

Table 1
Steam Review record variables.

Variable	Data type	Description
Review content	String	The textual information in the game reviews.
Recommendation flags	String	Whether reviewers recommend the game to others.
Playing hours	Numeric	The amount of time the reviewer has played as of the time the review was posted.
Reviewed items	Numeric	The number of games which have been reviewed by reviewer.
Helpful	Numeric	The number of people who regard the review as helpful.
Funny	Numeric	The number of people who consider the review to be amusing.

3.2. Data filtering

Data filtering consists of three stages. Firstly, duplicate review content data were eliminated. These duplicate reviews are likely to have been copied and contain ambiguous information, 20,835 data items remained. Then, 357 reviews that containing only punctuation and numbers were removed. It should be noted that swear words in Steam game reviews are automatically censored by the system and replaced by heart symbols [36]. Finally, a readability grade for each review was calculated using the Readability Consensus in the Textstat package.[5] Reviews with a readability grade less than 0th, which generally are too short to be informative (e.g., *'what [...] boyu'*), and those with a more than 200th readability grade, which contain repeat and spam information (e.g., *'UU [. . .] HH'*) are removed. After filtering, a new dataset with 20,000 reviews is obtained.

3.3. Qualitative and quantitative methods used in this study

3.3.1. Inductive content analysis (ICA)

In this study, we employ NVivo software[6] to perform ICA and identify the types of content contained in textual reviews. The inductive content analysis method is used to identify specific content or keywords from reviews at the word level [37]. When using NVivo, researchers first review the data to generate the initial categories (referred to as 'nodes' in NVivo). Then, the coding rules can be defined and the corresponding coding agenda can be created to reorganise and label the data based on these nodes and coding rules. When the text has been coded using this agenda, the initial definition of nodes and rules can be iteratively adjusted. Finally, substantive analysis can be performed on the basis of the optimised coding agenda in NVivo.

Subsequently, the initial nodes were recorded in condensed form, thus shaping the categories expressed in the reviews. After coding the text data and modifying the categories, the frequency of categories was calculated and the word frequency analysis and lemmatisation were also produced. In addition, any similar word between nodes was also clustered using the Pearson correlation coefficient (PPMCC) as the similarity metric.[7] In conclusion, input is made up of selected nodes and output from the software is a dendrogram and a summary of the PPMCC score. This approach enabled a comprehensive statistical analysis and interpretation of the idiomatic language used in Steam game reviews.

[5] https://pypi.org/project/textstat/

[6] https://support.qsrinternational.com/nvivo/s/article/How-do-I-cite-QSR-software-in-my-work

[7] https://help-nv10.qsrinternational.com/desktop/concepts/about_cluster_analysis.htm

3.3.2. Inductive thematic analysis (ITA)

As mentioned above, Steam reviews can contain multiple themes within a single review. This issue was addressed by generating multiple themes at the sentence level. Thematic analysis focusses on coding the theme of sentences; in other words, using the sentence as an analytical unit.

For example, let us consider a two-sentence review: *"GREAT, thrill of flying your own ship, and it has mods. 10/10 would play again"*. Using ICA analyses content at the word level. The review contains the following three nodes: Capitals, that is, words written in capitals; numerical rating, that is, a review score expressed numerically; jargon, that is, domain-specific vocabulary. The results of ICA for this example will extract *'GREAT'* from the node of Capitals, mods to Jargon, and *'10/10'* to numerical rating.

On the other hand, ITA works at the sentence level. The first sentence would be coded as a positive attitude towards the game, and the second sentence can be coded as a score theme. Similarly, these initial ITA results were used to define the themes that indicated the topics within the reviews. Once the stages are completed, it is possible to calculate, cross-analyse, and interpret individual themes using indirect references to background information about the game and the review recommendation flags.

3.3.3. LDA

In this study, the LDA topic modelling technique is used to summarise and identify the thematic structure of a large number of reviews. The assumptions of LDA for topic modelling are that texts with similar topics use similar groups of words, and latent topics can thus be found by looking for groups of words that frequently occur together, for example, in the NMS reviews on Steam. LDA is a generative model, and the parameters of the model are updated in an iterative procedure. The user needs to pre-set the number of topics for initial consideration. The items of text are then processed, assigning each word to one of the topics. Initially, these assignments are done randomly, so those random 'topics' may not make sense. In each iteration, the word-topic assignment is updated by calculating the probability that a specific word belonging to a topic is essentially a product of two probabilities. The first is the probability of a specific topic given a specific review $p(topic|review)$. For example, suppose that we consider four topics. One of the reviews may have the highest probability of belonging to topic four, though it also has a probability of belonging to the other three topics. The second is the probability of a specific word given the specific topic $p(word|topic)$. That is, a topic may generate a list of words with different probabilities. After a large number of iterations, the modelling can reach a steady state, that is, the values of the parameters of the probability distributions are roughly fixed. Then the assignments are acceptable, that is, we have each review assigned to a topic with a probability. We can also list the top number of words with the highest probabilities for each topic. Note that LDA is an unsupervised learning algorithm, which means that the (human) user needs to interpret these probability distributions as topics. Different numbers of topics may produce different results.

Pre-processed data and parameter settings are the main components impacting the performance of LDA-based topic modelling. Data preprocessing would be improved by adding special rules based on inductive content analysis of the lexical features of the review contents. To optimise the number of topics, this study recommends maximising the coherence measure (Cv) to ensure reasonable human interpretability for the LDA results. When a set of statements or facts is mutually supportive, they are considered coherent [38]. The topic coherence score was used to measure to what extent the model can be interpreted in a context. When the improvement of coherence stops showing a trend, the number of topics at that time comprises an optimised selection for LDA analysis. However, the supposition that the number of topics corresponding to when Cv reaches the peak is the most appropriate for the LDA model is not absolute. The parameters still require fine-tuning in accordance with the researcher's experience.

4. Experiment and results

We designed three experiments to investigate the contextual information on Steam.

4.1. Experiment I: Qualitative analysis using ICA

This experiment aimed to answer the question - what are the lexical features of reviews on Steam extracted using ICA? To manage the scale of the task, a small number of 200 samples (1%) were randomly selected from the complete list of 20,000 reviews. There were 18,111 words in the 200 samples, representing an average of about 90 words per review.

Each review was processed as an individual document by a human using NVivo. Based on the understanding of the data, we initialised categories, defined the coding rules, created a coding agenda, and then reorganised and labelled the data based on these categories and coding rules. When the text data were coded, we iteratively adjusted the initial definition of categories and rules. Finally, 1,084 nodes were generated and the relevant analysis was conducted in NVivo.

Table 2 shows the final coding rules. We then identified six main categories, 23 generic categories, and seven subcategories by manually applying these rules to the 200 samples. Table 3 summarises the final categories and their frequency among 18,111 words.

Table 2
Special coding rules for ICA.

Nodes	Description
Emoticons	Text containing emoticons
Missed symbol	Words missing necessary symbols
+ and - symbols	Text containing - or + symbols
Extra space	There are unexpected spaces between words
Missing space	Required space(s) are missing
Capitals	Words have been written in capitals
Repeated letters	There are letters repeated several times in a word
Numeric level	A number or fraction indicating a score
Currency	Currency symbols or numbers
Time	Words related to time
Pros and Cons	Text containing a listing of 'pros' and 'cons'
Game names	The name of another game
Developers	Words referring to the development team
NMS title	Words referring to NMS game
Game language in NMS	Special words used within NMS
Edited reviews	The review has been edited
Hyphen words	Words incorporating a hyphen punctuation mark
Jargon	Specialist video game vocabulary
'Not' contraction	Words such as don't (do not) and can't (can not)
Mixed language	Text contains more than one language
URL	Text is a URL
Computer configuration	Information about the reviewer's computer

Finding 1: Most of the numeric characters of the game review refer to time periods, currency and satisfaction levels. The largest generic category in Number relates to Time, where a unit of time, such as 'minute', 'hour' and 'year', usually follows the number, while numeric characters accompanied by currency symbols indicate that it is likely to be a price like $60 (NMS retail price). Moreover, we observed certain ratings patterns, for example, 10%, 4.5/5, and 10/10, indicating reviewers' level of enjoyment of the reviewer or the degree of appreciation of the game.

Finding 2: Game reviews contain slang terms on the Internet and abbreviations. 'TL;DR'(too long didn't read) as a common acronym was used to summarise long content, especially reviews with more than 190 words. Some typing shortcuts are used, such as 'u' for 'you' and 'tbh' for 'to be honest'. These abbreviations should be replaced with the entire word before capturing the topic or sentiment of reviews to avoid misunderstanding the result. Moreover, despite Steam engaging in automatic censorship and replacement of swear words before the publication of each review, some offensive language was abbreviated, which the censorship system cannot recognise.[8]

Finding 3: Many typos with multiple causes were identified in the game reviews. While many typos are spelling mistakes, more than 64, 5% of the samples were caused by no space between words and sentences. Finally, 42 words lacked punctuation, for example, 'don' and 'isn'.

Finding 4: Emoticons, capitalisations, and repeated letters are used to emphasise the emotions of reviewers. 7% of the reviews contain emoticons. Some reviewers prefer to repeat one particular letter in a word, such as 'nnnnnope' and 'waaay', or use the capitalisation of an entire word, such as 'EXTREMELY', to signal increased emotion, whether positive or negative.

Finding 5: Several language forms are used to describe the game's positive and negative aspects. Specifically, reviewers use 'pro' and '+' to refer to positive aspects and 'con' and '–' to refer to negative implications.

Finding 6: In game reviews, specific proper nouns are used, including the names of video games and jargon, with some parts written with abbreviations. The most commonly occurring game name is *Minecraft* (15.62%), and second is *Star Citizen* (4.69%). Certain players referred to abbreviated game names, for example, 'MC' for *Minecraft* and 'WOW' for *World of Warcraft*. Additionally, the names of the video games were not always formed using standard vocabulary.

Furthermore, jargon was often written as acronyms, for example, 'VR' (virtual reality), 'UI' (user interface), and 'FPS' (first-person shooter game or frames per second). Nevertheless, certain jargon was not presented as acronyms because they could neither be contracted nor substituted for alternative terms, for example, 'AAA' and 'modder'. Accordingly, when analysing data in the video game field, the name and jargon should be treated as proper nouns to avoid being misprocessed or providing incorrect contractions.

Finding 7: Some phrases expressing sarcasm in game reviews share linguistic similarities with the game name. To discover the diversity of expression, I collected different phrases referring to NMS, including 'NMS', 'no man' and 'no man's sky'. The language structure of almost one-third of the phrases with ironic meanings reflects that of the game titles in the sarcasm category, for example, 'one man's lie', 'no man's lie' and 'no man's loneliness.' Another sarcastic performance was related to the refund behaviour, such as the 'refund simulator'. This feature potentially inspired the identification of sarcasm in the game review study.

Finding 8: Game reviewers tend to edit their reviews as new patches are released. A cluster analysis demonstrated a similarity between the 'edited reviews' and the 'special game language in NMS' nodes. To explain this, we considered the frequency of the word and

[8] https://help.steampowered.com/en/faqs/view/6862-8119-C23E-EA7B

Table 3
Summary of the content analysis categories (Note that values in brackets show the number of words for each category).

Main category	Generic category	Subcategory
Symbols (80)	Emoticons (14) Missed symbols (42) + and- symbols (24)	
Space (204)	Extra spaces (9) Missing spaces (195)	
Letters (72)	Capital (69) Repeated letters (3)	
Number (98)	Numeric level (5)	
	Currency (37)	
	Time (56)	Bug time (5); Patch time (4); Playing hours (47)
Non-standard words (606)	Abbreviation (47)	
	Pros and Cons (6)	
	Game related (192)	Game names (40); Developers (61); NMS title (47); Game language in NMS (44)
	Edited reviews (19) Hyphen words (40) Informal words (23) Jargon (88) Misspelt words (65) 'Not' contraction (126)	
Language mode (24)	Mixed language (1) Sarcasm (13) URL (2) Computer configuration (8)	
Total: 1084 nodes		

found that words such as 'Patch', 'Next' and 'Origins' appeared in both categories. In fact, 'Next' and 'Origins' are the names of the patches for NMS. In addition, certain reviewers retained their original content and posts, instead of writing new reviews. The word 'original' (3.77%) indicated original content in an edited review. This mixed content likely contained contradictory opinions or attitudes within the same review.

4.2. Experiment II: Qualitative analysis using ITA

The ITA study aims to answer two questions: what are the commonly discussed topics of reviews on Steam, and what is the value of the recommendation element in Steam considering reviews' context and semantic features? This experiment used the same dataset as in Experiment I. There are approximately 1,130 sentences in the 200 samples, with an average of five sentences per review. Table 4 shows the coding rules. Table 5 presents the results of the classification of topics. At the sentence level, 1195 coded sentences were eventually categorised into five main categories. It should be noted that one sentence could belong to more than one node. These typically complex sentences include conjunctions, such as 'despite', 'although' and 'but', which are used to express various topics. We manually tagged a total of 17 generic topics and 46 subtopics. In Table 5, some of the categories identified are neutral with no emotional tendencies, such as Background and Player types; meanwhile, a part of the categories undoubtedly indicate that they will contain positive or negative sentiment, for example, judgement of feature value, reviewers of attacks and attitude. The feature value judgement category has the highest number of topics (details can be viewed in Fig. 1). Given that the lack of granularity could potentially decrease the sufficiency of building effective feature spaces, we associated this topic category with the ontological meta-model theory [39]. The framework of this meta-ontological model, comprising Physical, Structural, Communicational, and Mental layers, served as a foundation for displaying the detailed segmentation of game-related feature value judgements.

Fig. 1 shows the number of subcategories of the feature value judgement category within the main category related to the game in a stacked bar chart. Repetitiveness is the most common issue reported in the reviews. Game aspects such as being purposeful, third-person, exploration mechanics, and single-player gameplay are relatively entertaining to players. In contrast, negative views of saving progress, optimisation, and in-game environment (diversity) are comparatively frequent. The word frequency of NMS's disadvantages (Fig. 2(a)) indicates that game-related terms, such as 'planets', occur frequently. Combined with the original content, this provides evidence that reviewers used these terms to describe the game's high level of repetition.

Moreover, comparing the popularity of exploration mechanics and trading mechanics, exploration mechanics are acclaimed with an 87% positive tendency. In contrast, trading mechanics are considered unsatisfactory, with a positive trend of 36.4% (Fig. 1). This finding is supported by the frequency of the word related to the positive aspects of NMS (see Fig. 2(b)), in which the word 'exploration' commonly appears in this category. Meanwhile, words such as 'move' and 'number' in Fig. 2(a) contain aspects of trading mechanics.

Finding 9: Some features can be seen as positive and negative, including repetitiveness, diversity, aimlessness, and purposefulness. Not surprisingly, different reviewers perceive some features differently, so the same feature can be regarded positive by some reviewers and negative by others. In particular, a reviewer was observed to express that their favourite characteristic was aimlessness, which was therefore classified as aimless (positive) instead of purposeful.

Finding 10: The recommendation flag of a review did not always correspond to the sentiment of the reviewers. Table 6 compares the difference between the recommendation flags on the Steam page, and the recommendation classifications based on the reviews' language. A total of 48 reviews contain sentences that directly mention whether they would recommend the game. Two cases clearly express a non-recommendation intent in the comments, while their recommendation flags are 'recommend' (e.g., *'more like no man buyed it [. . .]', 'STAY AWAY FROM [. . .]'*). This might indicate that reviewers have assigned the wrong recommendation labels considering the content of their reviews. A further notable result in one case is a sentence that represents how the reviewer does not want to offer a recommendation to others. However, they have decided to do so due to Steam's mechanics.

Table 4
Special coding rules for ITA.

Nodes	Description
Background	Sentences contained background information about NMS.
Patches	Sentences related to patches' information.
Feature value judgement (Negative/Positive)	Sentence contained a clearly negative/positive description of game's aspect or content.
Comparison	Sentence contained a comparison between NMS and other games.
Game length	Sentence contained topics discussing the length of the game.
Score	Sentence contained topics discussing the score that NMS should receive.
Attack reviewers	Sentence contained information that attacked other reviewers.
Player types	Sentence described different types of players.
Target player	Sentence contained an explanation that the game is only recommended to particular groups of players.
Early access	Sentence mentioned any information related to early access.
Refund	Sentence contained information that the reviewer wanted or had already received a refund.
Streaming	Sentence mentioned streaming or YouTubers.
Emotional attitude	Sentence contained information that in relation to the developer or game.
Recommendation	Sentence mentioned whether they will recommend the game to others.

Table 5
Summary of the thematic analysis categories (Note that values in brackets show the number of sentences for each category).

Main category	Generic category	Subcategory
Game related (552)	Background (76)	General information (13); Promotion (63)
	Patches (47)	Bug fixed (7); Encouragement (38); Writing review for it (2)
	Feature value judgement: (Negative 234; Positive 136)	Physical layer (52); Structural layer (170); Communicational layer (40); Mental layer (108)
	Comparison (32)	
	Game length (1)	
	Score (3)	
Player related (56)	Attack reviewers (6)	
	Player types (27)	Non-pre-ordered player (7); Pre-ordered player (9); Returned player (11)
	Target gamer (23)	
Platform related (99)	Early access (11)	
	Retail price (69)	Not worth price (41); Worth price (14); Suggested price (14)
	Refund (18)	
	Streaming (1)	
Attitude (487)	For Developer (92)	Like (62); Dislike (30)
	For Game (341)	Like (149); Dislike (80); Natural (12)
	Recommendation (54)	Recommend (22); Not recommended (22); No suggestion (1)
Others (1)		
Total: 1195 nodes		

Table 6
Results of the cross-tabulation analysis concerning ITA's recommendation and Steam's recommendations.

		Steam	
		Not Recommend	Recommend
ITA	Not Recommend	23	2
	Recommend	0	22
	No recommended suggestion	0	1

Furthermore, we observed that the themes of emotional expression included in the reviews are primarily divided into two types: one concerns the game itself, while the second concerns the developer (Table 7). 184 reviews mentioned the attitudes of the reviewers towards the game, and 63 reviews referred to their attitudes towards the developer. Of course, some of these reviews could touch on both topics at the same time. Regarding the emotions of the game, among those who enjoyed the game, 92 selected to recommend the game to others, while 15 did not. There are 14 cases of reviews showing a willingness to recommend the game while still including sentences that clearly conveyed the reviewer's dislike of the game. The attitudes of reviewers towards developers are similar to their attitudes towards the game, which means that the majority of reviewers recommend the game and like its developer and vice versa. However, the emotional tendency of the language and the correlation with recommendation flags are less apparent than the results in Table 6. Therefore, I may not conclude a definite causal relationship between like or dislike and a recommendation.

Finding 11: The relationship between recommendation flags and potentially negative topics is not significant. To further evaluate the relationship between reviews' sentiments and recommendation flags, we investigated the distribution of topics that could potentially

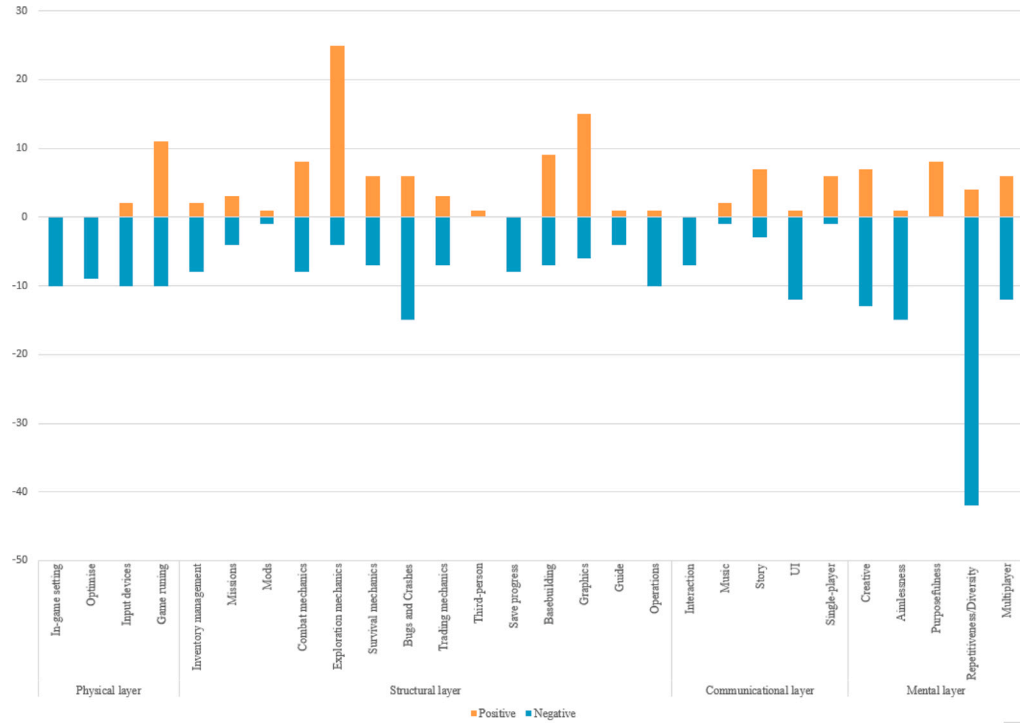

Fig. 1. Details of positive and negative counts for each individual category in feature value judgements (note that orange bars indicate positives; blue bars indicate negatives).

Fig. 2. (a) Word cloud visualising texts relating to positive items. (b) Word cloud visualising the contents relating to negative items.

Table 7
Results of the cross-tabulation analysis concerning emotional attitudes and recommendations.

		Steam	
		Not Recommend	Recommend
ITA	Dislike (Game)	54	14
	Like (Game)	15	92
	Neutral (Game)	3	6
	Dislike (Developer)	18	7
	Like (Developer)	4	34

contain negative sentiments and their recommendation flags. Table 8 shows the results. It is apparent from this table that very few reviews were labelled as recommended if they included the topic of refund (14.29%). Nevertheless, despite the reviewers complaining that NMS was overhyped and engaging in comparisons with other games, this did not affect about half of them recommending the game. The topic of attacking or disagreeing with other reviewers was found to be associated with the recommended group in the NMS reviews. This topic's emotions are complex; they indicate negative attitudes towards other reviewers, but also imply potentially positive attitudes towards the game.

4.3. Experiment III: Topic modelling using LDA

This experiment aims to further investigate the semantic features of Steam reviews using LDA and compare its results with those obtained from Experiment II using ITA. This experiment used a larger sample that included all 20,000 cleaned reviews.

Before analysis, the data were cleaned and preprocessed according to two key activities, the first of which required the sentence tokenizer

Table 8
Results of the cross-tabulation analysis among potential negative topics and recommendations.

		Steam	
		Not Recommend	Recommend
ITA	Refund (14)	85.71%	14.29%
	Comparison (20)	50%	50%
	Hype/Promotion (40)	55%	45%
	Target player (18)	22.22%	77.78%
	Attack other reviewers (4)	0%	100%

from Natural Language ToolKit (NLTK[9]) to be applied to divide reviews into sets of sentence-level data. For each sentence, the text was processed with lowercase conversion and expanding contractions, and then several rules were executed to clean the data, including correct spelling, replacing emoticon, replacing or removing numbers, removing punctuations, stemming and lemmatisation, removing URL, removing tags,[10] and removing stopwords. The programme also removes the most common words ('game') and the least frequent words ('boooooo') to eliminate low-level information. Subsequently, the construction of the bigram and trigram models using NLTK's NgramAssocMeasures to complete the word tokenizer and identify the phrases within the data.

Furthermore, a modification of these standard rules was carried out to achieve a more rational data preprocessing programme, as well as to reflect the particular linguistic characteristics of Steam reviews. Specifically, regular expression operations (RE) were used to replace specific numbers and symbols, for example, converting '< 3' to 'heart' and so forth. Meanwhile, RE was adopted to expand contraction-abbreviated proper nouns, for instance, 'hrs' and 'f2p'. Using these special regular expression rules, we unify various expressions of the same thing by different users, thereby enhancing topic-modelling accuracy. Furthermore, the overall performance of the spelling correction was improved using the PySpellCheck[11] customised dictionary, which was modified to incorporate game-related terms into the vocabulary.

In this study, the coherence measurement for different numbers of topics was used to optimise the number of topics. We initially varied the number of topics from 1 to 24 with an increment of 1. The highest coherence corresponded to 10 topics. Having compared the results for the various topics, we determined to set 10 topics overall, in order to obtain the strongest result. Although more topics may prove beneficial as a means of reflecting the diversity of reviews, overly repetitive words in the same subtopics would be unnecessary and hinder reasonable human interpretation.

Table 9 shows the first ten topics and their associated words. Topics 1, 6, 7, and, to some extent, topic 3 focused on the game contents. Compared with the other topics, topics 4, 8, and 10 are relatively negative themes related to reporting game problems. Unlike topic 6, which notes destructive game items, these topics were not related to game resources. Moreover, topic 3 comprises more words discussing the game's positive aspects; topic 5 five plus a portion of topics 2 and 9 concern themes related to the game developer. Since reviewers express positive attitudes towards patches and a group of players indicated they 'forgive' NMS developers, we speculate that some of the information on this topic is positive.

Combining its results with the classification obtained from Experiment II using ITA, topics 2, 4, 8 and 10 were observed in both experiments. Topics 4 and 8 were classified as subcategories in Experiment II referring to the internal feature judgement of the video game. In contrast, topics 2 and 10 are categorised as generic categories in the previous experiment, as they are relevant to the economic themes of the third-party distribution platform and the additional game contents.

5. Discussion and conclusion

One of the reasons that the performance of NLP techniques in processing game-related data is unsatisfactory is that the linguistic features of game reviews are different from other corpora. In this study, we performed a comprehensive analysis to begin to understand the characterising and linguistic features of game reviews through an initial case study that looked at NMS reviews on Steam.

Our results align with previous studies [11,36] in highlighting the prevalence of features such as numerical ratings and emoticons in game reviews, reflecting their importance in shaping user interactions. We identified that game reviews often contain typos, internet language, and jargon, which could present challenges for NLP-based models, as noted in previous research [22]. Our findings highlight sarcasm as a significant feature in game reviews, particularly through puns derived from the game's name.

Although a single review could contain discussions of both the game's advantages and disadvantages, this study shows that additionally, reviews can contain both original and edited content, which may provide mixed content and opposing perspectives. While edited reviews pose challenges for automated language analysis, they also provide valuable insights into shifts in reviewers' opinions and evolving attitudes.

We found that the themes identified by ITA, such as crash and bugs fixed via updates, substantially overlapped with those identified in [21]. Interestingly, given the connection between the element of recommendation and the sentiment of the reviews, one finding is that there is limited direct evidence that a player's decision to recommend the game reflected their review's emotional attitude, despite reviewers tending not to recommend the game if their review contained negative emotions interpreted towards the game or its developers. This finding is in contrast to previous studies [5,12,27,28] which straightforwardly treated recommendation flags as sentiment labels. Although there is some connection between recommendation and sentiment, more research is needed to confirm the relationship, especially regarding negative emotional content. A further finding was that reviewers may express various emotions towards several different objects. Therefore, a successful analytic model must distinguish the various objects to which a sentiment can refer to precisely measure the emotional feedback of the reviewer.

Furthermore, we examined the performance of topic extraction using the LDA algorithm. This study presented a qualitative evaluation of the language characteristic of Steam game reviews, incorporating qualitative analysis results into the establishment of the NLP system. The overall topic classification using LDA provided useful insights, with topics such as enjoyment, patches, refund, promotion, bugs, game content, and good in-game elements aligning with those identified in Lu et al.'s [4] research on NMS.

This study has the following limitations. First, given that this study selected NMS as a special case study, it would be fruitful if future research incorporated additional reviews about other games so that the findings of this study can be generalised to other games. Second, while readability was explored, more research is needed to draw more definitive conclusions on this topic. Additionally, as LDA is an unsupervised topic model, results and interpretations may vary depending on input parameters.

In conclusion, this study offered insights into the types of information and structural patterns present in NMS player reviews on Steam. Our results provided a detailed understanding of these reviews, focussing on language performance and content semantics. This study used both manual and NLP analysis approaches to highlight the differences between human and automated analysis.This research offered information to researchers and video game practitioners on the characteristics of game reviews. We believe that the findings of this research offer a valuable foundation for processing review data and applying NLP analysis to game reviews on Steam.

[9] https://www.nltk.org/
[10] https://beautiful-soup-4.readthedocs.io/en/latest/
[11] https://pypi.org/project/pyspellchecker/

Table 9
Example of top 5 words generated under 10 topics.

Topics	Summary	Cluster words
1	Game contents: exploring the space and building the base.	‘planet’, ‘ship’, ‘space’, ‘system’, ‘base’
2	Patches: incorporation of new game contents and the patches’ names.	‘update’, ‘release’, ‘launch’, ‘next’, ‘add’
3	Good elements in-game: NMS’s most significant mechanics based on the previous analysis.	‘space’, ‘exploration’, ‘explore’, ‘universe’, ‘great’
4	Game running issues: ‘fps’ refers to frames per second based on the context.	‘run’, ‘fps’, ‘issue’, ‘personal_computer’, ‘setting’
5	Developers: related to the various appellations of NMS’ developer team.	‘hello_games’, ‘promise’, ‘developer’, ‘sean_murray’, ‘deliver’
6	Bad elements in-game: related to reviewers’ complaints about repetition in NMS.	‘planet’, ‘different’, ‘nothing’, ‘boring’, ‘resource’
7	Enjoyment: related to the reviewer’s positive game experience.	‘play’, ‘fun’, ‘enjoy’, ‘love’, ‘friend’
8	Bugs: complaints relating to the game’s bugs and crashes.	‘time’, ‘bug’, ‘crash’, ‘save’, ‘bad’
9	Promotion: related to the evaluation of NMS’ game promotion.	‘people’, ‘hype’, ‘expect’, ‘lie’, ‘trailer’
10	Refund: pertained to refund behaviours and certain negative assessments of the game.	‘refund’, ‘steam’, ‘buy’, ‘lie’, ‘garbage’

CRediT authorship contribution statement

Xinge Tong: Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Data curation, Conceptualization, Formal analysis. **Ian Willcock:** Supervision, Writing – review & editing, Conceptualization. **Yi Sun:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

We are writing to confirm that we have no conflict of interest related to my submission to your journal. The article that we are submitting, titled “What is A Game Review: A Case Study Approach to Defining Player Reviews” was conducted with no external funding or support from any organisation or individual that could potentially influence the findings or conclusions of my research.

We can assure you that our research was conducted with the highest level of scientific rigor and that the findings are based solely on the data collected during the experiments. We have taken care to provide a clear and unbiased interpretation of the results, without any external influence.

We believe that full transparency and disclosure of potential conflicts of interest is essential to maintaining the integrity of the research process, and we are committed to upholding the ethical standards of the research community.

Data availability

The data that has been used is confidential.

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